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EEGdiff / DiffEEG — Glossary of Terms

A reference for the terminology used throughout this project (the DiffEEG paper, the processing scripts, and the evaluation ladder). Grouped by theme.

1. Evaluation / adaptation settings

These describe *how much* (if anything) we train on top of the frozen diffusion backbone. Ordered from “no training” to “most training”.

Zero-shot transfer

Take a classifier that was **already trained on another dataset/task** and apply it to a **new** dataset with **no further training**. - Uses **no** labels from the target dataset. - Only works when the label space matches (e.g. seizure → seizure). - In our work: reuse the TUSZ-trained seizure head on Bonn / Siena (Bonn ROC-AUC 0.716, Siena 0.626).

Parameter-free / non-parametric probe (k-NN)

Classify each point by its **nearest neighbours** in the frozen embedding space. A classifier with **no learned weights at all** (nothing is trained). - Uses the target dataset’s **own labels** — the training embeddings form the neighbour bank — but performs **no gradient training**. - **NOT** the same as zero-shot: zero-shot uses no target labels; k-NN does. - Measures the *local geometry* / clustering of the embedding. - In our work: 5-fold CV balanced-accuracy k-NN (k=10) on the 4800-d embedding (**tab:knn**, and the t-SNE panel annotations).

Linear probing (a.k.a. head-only)

Freeze the backbone, attach a **new** classifier head, and **train only that head** on the target dataset’s training split. The backbone never updates. - “Linear” is used loosely: our head is a small **2-layer MLP** (4800 → 512 → 256 → K, BatchNorm + GELU + dropout), not a single linear layer. - Isolates the quality of the frozen representation with a small trained readout. - In our tables: the “Head only, no RL” rows.

Partial fine-tuning (unfreezing)

Unfreeze **some** of the backbone (from the deepest encoder level up to the whole encoder) and train it together with the head, using a gentler learning rate on the backbone (`_backbone = 0.01 · _head`). - Flags: `--unfreeze last_level | last_two_levels | encoder_all | all`.

Full fine-tuning

Unfreeze the **entire** backbone (`--unfreeze all`) and train end-to-end.

RL decision head (REINFORCE)

An add-on classification head trained with a policy-gradient (REINFORCE) objective whose reward is **macro-F1**, to directly optimise the imbalanced-class metric rather than cross-entropy. Combined with any of the above (e.g. “Unfrozen + RL”). Tried PPO as an alternative — did not beat REINFORCE.

One-line summary: zero-shot (reuse a foreign classifier) → k-NN (no training, target labels only) → linear probing (train a head) → partial FT → full FT (+ optional RL head).

2. Metrics

Accuracy (plain)

Fraction of correctly classified samples. **Misleading under class imbalance** (a majority-only predictor scores high).

Balanced accuracy

Mean of the **per-class recall** (= macro-averaged recall). Robust to imbalance; a majority-only predictor scores at chance (= $1 / \text{\#classes}$). This is what CBraMod / LaBraM / REVE report, and what we report to compare with them. - **Balanced accuracy (the metric)** **balanced dataset (subsampling)**. The foundation-model baselines use the **full imbalanced data** and report the balanced-accuracy metric — they do **not** subsample to balance the data.

Macro-F1

Unweighted mean of the per-class F1 scores (each class counts equally). Imbalance-robust; good for multiclass tasks with rare classes.

Weighted-F1 (W-F1)

Mean of per-class F1 weighted by class support. Closer to plain accuracy; dominated by the majority class.

ROC-AUC (AUROC)

Area under the ROC curve (true-positive vs false-positive rate). **Invariant to class prevalence** — it does not change if you resample the negative class.

AUC-PR / PR-AUC (Average Precision)

Area under the precision–recall curve. **Strongly depends on prevalence** — evaluating on a balanced test set inflates it. (This is the confound behind the “effect of test-set class balance” section: same model, AUPRC rises from 0.48 at 6.7% seizure prevalence to 0.86 at full balance, while AUROC is unchanged.)

Cohen’s Kappa ()

Agreement between predictions and labels **corrected for chance**. Used for multiclass tasks (e.g. TUEV) and inter-rater reliability.

Sensitivity / Specificity / Precision / Recall

- **Sensitivity = Recall** (of the positive class) = $TP / (TP + FN)$ — ability to catch positives (e.g. seizures).
- **Specificity** = $TN / (TN + FP)$ — ability to correctly pass normals.
- **Precision** = $TP / (TP + FP)$ — of the predicted positives, how many are real.

Chance level

The score a trivial/random classifier gets: $1 / \text{\#classes}$ for balanced accuracy and accuracy; 0.5 for ROC-AUC.

3. Data splits (how train/test are separated)

Subject-wise / subject-independent split

Each **subject** appears in **only one** of train / test. Measures genuine generalisation to new people. The honest, harder setting for within-subject datasets (SEED-V, EEGMMIDB, Mumtaz).

Patient-wise split

The clinical version of subject-wise: no **patient** appears in both sides (TUSZ, TUAB, Siena). The official TUH corpus splits are patient-disjoint by construction.

Clip-wise / trial-wise split (CBraMod 5:5:5)

Split by **stimulus clip** / **trial** within each session, so **all subjects appear in every split**; only the movie clips differ. This is how CBraMod / LaBraM / REVE evaluate SEED-V and (via EC+EO windows) Mumtaz — *subject-dependent* but clip-disjoint. Must match this to compare against their leaderboard numbers.

Pooled random split (leaky / within-subject diagnostic)

Pool all windows and draw a fresh random train/test split, **ignoring** the official subject split — the same subject (and often the same recording) leaks into both sides. Inflates results; used only as a diagnostic to show how much of a gap is due to the split vs the model.

Prevalence-matched evaluation

Re-score the **same** model probabilities after subsampling the negative class to a target positive-class fraction (e.g. 6.7% natural vs 29% “balanced” as used by seizure-FM benchmarks). Reveals how much of an AUPRC gap is a test-set-balance artefact rather than a real capability difference.

4. Model & pipeline concepts

Diffusion backbone

DeepEnhancedEEGDiffusionModel — a 1-D U-Net (40.7 M params) pretrained with a **denoising-diffusion** objective (predict the noise added to EEG at a random timestep) on non-seizure clinical EEG. No labels are used in pretraining (self-supervised / generative).

Multi-timestep diffusion-probing embedding

To turn the generative backbone into a feature extractor: run the frozen encoder at 5 diffusion timesteps (50, 250, 500, 750, 950), global-average-pool each encoder level, and concatenate $\rightarrow$ a fixed **4800-dimensional** embedding ($64 \cdot (1+2+4+8) \cdot 5$), independent of the input length.

Channel harmonization (22-channel canonical frame)

Map any dataset’s electrodes onto our fixed 22-channel 10–20 frame **by name** (select — never average or interpolate, because the backbone’s filters are electrode-position specific). Missing electrodes are **zero-filled**; first-match-wins so a real electrode is never overwritten by an alias. - Canonical order: FP1, FP2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T3, T4, T5, T6, A1, A2, FZ, CZ, PZ, ROC.

Normalization (own stats, z-scoring)

Per-channel z-scoring using **each dataset’s own** mean/std (computed over its train split). Makes the physical unit irrelevant (volts vs μ V) — never reuse another dataset’s stats, which would crush the signal toward zero.

Windowing

Cut continuous EEG into fixed **5-second / 1280-sample windows at 256 Hz** (the backbone’s native input). Non-overlapping unless a shorter stride is used to increase sample count.

Common-average reference (CAR)

Re-reference each electrode by subtracting the mean across the present scalp electrodes. A standard preprocessing step (used by CBraMod/REVE); we apply it in the “matched” Mumtaz pipeline.

Band-pass / notch filter

- **Band-pass** (e.g. 0.1–30 Hz): keep the physiologically informative band, remove slow drift and high-frequency EMG/noise.
 - **Notch** (50 or 60 Hz): remove power-line interference.
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5. Quick reference — what we report per task type

Task type	Primary metrics
Binary clinical (TUAB, TUSZ, Siena, Bonn, Mumtaz)	Balanced acc, ROC-AUC, AUC-PR
Multiclass events / sleep (TUEV, Sleep-EDFx)	Balanced acc, macro-F1, Cohen’s
Motor / emotion (EEGMMIDB, SEED-V)	Balanced acc, macro-F1

Balanced accuracy is the common comparison metric against foundation-model baselines (CBraMod, LaBraM, REVE, NeurIPT, BIOT).